

Introduction to Quantum Computing

From Classical Bits to Quantum Algorithms

2026 Edition

An Independent Modern Textbook
Research cutoff: September 2026

Abstract

This textbook provides a mathematically careful yet accessible introduction to quantum computation and quantum information for readers who know basic algebra, trigonometry, and complex numbers but need not already know linear algebra, quantum mechanics, or quantum programming. It develops classical information and computation, reversible logic, single- and multi-qubit quantum theory, measurement, the Bloch sphere, linear-algebraic tools, entanglement, quantum circuits, error correction, protocols, and foundational quantum algorithms through Shor's factoring algorithm. A companion interactive website synchronizes notation, examples, and laboratory exercises with this PDF. Material labeled with the research cutoff reflects the state of the field as of September 2026 and should be re-checked as hardware and software evolve.

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How to Use This Book

Each chapter in the PDF corresponds to a section of the interactive web textbook at the project repository. When you see an **Interactive Lab** box (the `labbox` environment), open the indicated lab at `/playground/<lab-id>` on the companion website to manipulate states, run simulations, and verify calculations step by step.

Chapter 1

Classical Information and Computation

Before quantum mechanics enters the story, we need a precise language for *classical* information: how bits encode data, how logic gates transform bits, how arithmetic is built from gates, and how classical computers are modeled mathematically. This chapter establishes the baseline against which quantum advantages are measured. Every concept here reappears—often in generalized form—when we study qubits, reversible circuits, and complexity classes such as **BQP**.

1.1 States and Information

1.1.1 Physical encoding

Physical systems can encode information by occupying distinguishable states. A switch may be open or closed; a magnetic domain may point up or down. The abstraction that captures this binary distinction is the *classical bit*: a variable $b \in \{0, 1\}$.

For n independent bits, a register holds one of 2^n distinct strings. If each bit is equally likely and independent, the number of configurations is

$$N = 2^n. \tag{1.1}$$

For $n = 3$, the eight strings are 000, 001, 010, 011, 100, 101, 110, 111. Bits may represent integers (via binary encoding), characters (via ASCII or Unicode), machine instructions, or any digital datum once a convention is fixed.

1.1.2 Entropy and distinguishability

In Shannon’s theory, a uniform random n -bit string carries n bits of entropy. Distinguishability is absolute: observing a bit yields a definite 0 or 1 with no residual uncertainty unless the physical medium is noisy—a theme that returns in Chapter 5 when we protect quantum information.

Interactive Lab 1: Binary State Explorer

Navigate to `/playground/binary-states` on the companion website. Select register sizes $n = 1, 2, \dots, 8$ and verify that the state count matches (1.1). Toggle individual bits and observe how each flip changes the decimal interpretation of the string.

Example. *How many distinct 10-bit registers exist? Using (1.1) with $n = 10$,*

$$N = 2^{10} = 1024.$$

This is exactly one kibibit (1 KiB) of addressable classical states when each address is one byte-aligned block of ten bits.

1.2 Binary Numbers

1.2.1 Positional notation

Positional notation interprets a bit string $b_{n-1}b_{n-2}\cdots b_0$ with $b_j \in \{0, 1\}$ as the integer

$$(b_{n-1}\cdots b_0)_2 = \sum_{j=0}^{n-1} b_j 2^j. \quad (1.2)$$

The subscript 2 reminds us that digits are base-two. Leading zeros do not change value: $00101_2 = 101_2 = 5_{10}$.

1.2.2 Conversion and arithmetic preview

To convert from decimal to binary, repeated division by two extracts remainders as least-significant bits. Addition in binary uses the same positional rules as decimal, but carries propagate when a column sums to 2 or more.

Example. *Evaluate 1011_2 in decimal:*

$$1011_2 = 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 8 + 0 + 2 + 1 = 11_{10}.$$

1.2.3 Fixed-width registers

Real hardware uses fixed-width registers. An n -bit unsigned integer wraps on overflow modulo 2^n . Two's-complement encoding represents signed integers; the most significant bit indicates sign. Quantum algorithms often treat bit strings as labels for computational basis states $|b_{n-1}\cdots b_0\rangle$, making binary conventions essential.

1.3 Logic Gates

1.3.1 Boolean variables and truth tables

Boolean variables $A, B, C \in \{0, 1\}$ combine through *logic gates*. Table 1.1 lists fundamental gates; each gate is a function $\{0, 1\}^k \rightarrow \{0, 1\}$ for some k .

Table 1.1: Elementary logic gates and their outputs.

Gate	Symbol	Definition
NOT	$\overline{A}$	$\overline{0} = 1, \overline{1} = 0$
AND	$A \wedge B$	1 only if both inputs are 1
OR	$A \vee B$	1 if at least one input is 1
XOR	$A \oplus B$	1 if inputs differ
NAND	$\overline{A \wedge B}$	negated AND

XOR is *linear* over $\mathbb{F}_2$: $A \oplus B \oplus C$ is the parity of three bits. It is also its own inverse: $(A \oplus B) \oplus B = A$.

Interactive Lab 2: Logic Gate Simulator

At </playground/logic-gates>, build circuits from NOT, AND, OR, and XOR. Verify De Morgan identities interactively by comparing truth tables of equivalent networks.

1.3.2 Universal classical sets

A gate set is *universal* if any Boolean function can be built from it. {NAND} alone is universal: NOT is $\text{NAND}(A, A)$, AND is $\text{NAND}(\text{NAND}(A, B), \text{NAND}(A, B))$, and OR follows from De Morgan. Reversible quantum gate sets play an analogous role in Chapter 4.

1.4 Boolean Algebra

1.4.1 Axioms and identities

Boolean algebra axiomatizes logic with operations $\wedge, \vee, \neg$. Key identities include

$$A \wedge 1 = A, \quad A \vee 0 = A, \quad (1.3)$$

$$A \wedge 0 = 0, \quad A \vee 1 = 1, \quad (1.4)$$

$$A \wedge \bar{A} = 0, \quad A \vee \bar{A} = 1. \quad (1.5)$$

Associativity, commutativity, and distributivity hold as in ordinary algebra, with the caveat that $A \wedge (B \vee C) = (A \wedge B) \vee (A \wedge C)$.

1.4.2 De Morgan's laws

De Morgan's laws relate negation and conjunction/disjunction:

$$\overline{A \wedge B} = \bar{A} \vee \bar{B}, \quad \overline{A \vee B} = \bar{A} \wedge \bar{B}. \quad (1.6)$$

These identities underpin NAND-based synthesis and appear in circuit optimization.

Example. Simplify $\overline{A \wedge (B \vee C)}$ using (1.6):

$$\begin{aligned} \overline{A \wedge (B \vee C)} &= \bar{A} \vee \overline{B \vee C} \\ &= \bar{A} \vee (\bar{B} \wedge \bar{C}). \end{aligned}$$

1.4.3 Normal forms

Any Boolean function admits a *sum-of-products* (disjunctive normal form) representation as an OR of AND clauses. Minimization—finding the fewest gates—is NP-hard in general, foreshadowing complexity themes in Section 1.8.

1.5 Adders

1.5.1 Half adder

A *half adder* adds two bits without incoming carry, producing sum S and carry-out C :

$$S = A \oplus B, \quad C = A \wedge B. \quad (1.7)$$

1.5.2 Full adder and ripple carry

A *full adder* includes carry-in C_{in} :

$$S = A \oplus B \oplus C_{\text{in}}, \quad (1.8)$$

$$C_{\text{out}} = (A \wedge B) \vee (C_{\text{in}} \wedge (A \oplus B)). \quad (1.9)$$

Chaining n full adders yields an n -bit ripple-carry adder. Worst-case delay is $O(n)$ gate layers because the carry must propagate from least to most significant bit—a *depth* bottleneck that quantum adder designs also confront.

Interactive Lab 3: Binary Adder

Use `/playground/binary-adder` to add multi-bit numbers and trace carries step by step. Compare ripple depth with your hand calculation of (1.8)–(1.9).

Example. Compute $1101_2 + 1011_2$:

$$\begin{array}{r} 1101_2 \text{ (13}_{10}\text{)} \\ + 1011_2 \text{ (11}_{10}\text{)} \\ \hline 11000_2 \text{ (24}_{10}\text{)}. \end{array}$$

Column-wise: $1 + 1 = 0$ carry 1; $0 + 1 + 1 = 0$ carry 1; $1 + 0 + 1 = 0$ carry 1; $1 + 1 + 1 = 1$ carry 1.

1.6 Reversible Computation

1.6.1 Information loss in irreversible gates

A gate is *reversible* if outputs uniquely determine inputs. NOT and XOR (with one output wired as copy) are reversible; AND is not—knowing output 0 does not distinguish $(0, 0)$ from $(0, 1)$. Landauer’s principle links irreversible erasure to minimum heat dissipation, motivating reversible architectures.

1.6.2 Toffoli and Fredkin gates

The *Toffoli* (controlled-controlled-NOT) gate maps $(a, b, c) \mapsto (a, b, c \oplus ab)$. With $c = 0$, it computes AND reversibly into the third wire. The *Fredkin* (controlled-SWAP) gate conditionally exchanges two target bits. Any classical computation can be embedded in a reversible circuit using ancilla bits initialized to 0.

Example. To emulate $c = a \wedge b$ reversibly, start with $(a, b, 0)$ and apply Toffoli to obtain $(a, b, a \wedge b)$. The original AND output is read from the third wire; (a, b) remain available for later use.

Quantum evolution before measurement must be unitary (hence reversible). Classical reversible logic is the direct template for quantum circuit synthesis.

1.7 Classical Error Correction

1.7.1 Repetition code

Noise flips bits with small probability p . The three-bit *repetition code* encodes logical 0 as $0_L = 000$ and logical 1 as $1_L = 111$. A single bit flip yields a word at Hamming distance 1 from one codeword and distance 2 from the other; *majority vote* decoding corrects one error.

1.7.2 Parity checks

Parity $p = a \oplus b$ summarizes two bits in one check bit. Longer block codes add multiple parity checks to locate errors via *syndromes*. The quantum three-qubit bit-flip code (Chapter 5) is a direct quantum analogue.

Example. Codeword 101 suffers a flip on the middle bit, becoming 111. Majority vote: two ones $\Rightarrow$ decode as 1. The syndrome “even number of ones in first two bits” would flag an error in more elaborate schemes.

1.8 Computational Complexity

1.8.1 Asymptotic growth

Algorithm cost is measured as a function of input size n . Common classes include $O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$, and $O(2^n)$. Exponential growth renders problems intractable for large n on any physically realizable classical machine.

1.8.2 P, NP, and BQP preview

P is the class of decision problems solvable in polynomial time on a deterministic Turing machine. **NP** contains problems whose *yes* certificates are verifiable in polynomial time; NP-complete problems are the hardest in NP. Quantum computers do *not* automatically solve NP-complete problems in polynomial time.

BQP (bounded-error quantum polynomial time) is the quantum analogue of P: problems solvable by a quantum computer with error probability $< 1/3$ in polynomial time. Factoring (Shor) and unstructured search (Grover) illustrate problems where quantum algorithms offer provable advantages over known classical methods for specific tasks.

1.9 Turing Machines

1.9.1 The model

A *Turing machine* consists of an infinite tape, a head that reads/writes symbols, a finite state controller, and a transition function. At each step the head reads a symbol, writes a symbol, moves left or right, and updates state. Despite its simplicity, the model captures what is *classically computable*.

1.9.2 Church–Turing thesis

The Church–Turing thesis asserts that any physically realizable *classical* computation can be simulated by a Turing machine with at most polynomial overhead. Quantum computation extends the physical model: the open question is whether all quantum processes admit efficient classical simulation. Evidence from Shor’s algorithm suggests the answer is no for some problems.

Example. *A Turing machine adding two n -bit binary numbers can scan the tape once per bit with carry, using $O(n)$ steps and $O(n)$ tape cells—polynomial in the input size, hence membership in P for this functional problem.*

Exercises

1. How many distinct states does a 12-bit register have? Express the answer as a power of two and as a decimal integer.
2. Using De Morgan’s laws, rewrite $\overline{(\overline{A} \vee B) \wedge C}$ without nested negations above the variable level.
3. Add $1110_2 + 0111_2$ using ripple carry. How many column carries occur?
4. Prove that XOR is associative: $(A \oplus B) \oplus C = A \oplus (B \oplus C)$.
5. Explain why AND alone cannot form a NOT gate without constant inputs.
6. In the three-bit repetition code, what is the minimum Hamming distance between valid codewords? How many bit errors can be corrected?
7. Classify each function as constant or balanced for $n = 1$: $f(x) = 0$, $f(x) = 1$, $f(x) = x$, $f(x) = \overline{x}$.
8. In one paragraph, relate reversible computation to the requirement that quantum gates be unitary before measurement.

Chapter 2

The Qubit

A qubit generalizes the classical bit. Whereas a classical bit is deterministically 0 or 1, a qubit may occupy a *superposition*—a normalized linear combination of $|0\rangle$ and $|1\rangle$ with complex coefficients. This chapter develops the state-space picture, measurement rules, alternative bases, phase, the Bloch sphere, and elementary gates that manipulate single-qubit states.

2.1 What Is a Qubit?

2.1.1 State vector representation

A pure one-qubit state is a unit vector in a two-dimensional complex Hilbert space:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (2.1)$$

In the *computational* or *Z basis*,

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (2.2)$$

2.1.2 Superposition versus classical randomness

A qubit is *not* “both 0 and 1 at once” in a classical sense. Amplitudes α, β carry *magnitude* and *phase*. Two states with identical probability distributions for *Z* measurements may differ in other bases—a distinction with no classical analogue. Interference arises when amplitudes combine before measurement.

Interactive Lab 5: Single-Qubit State Explorer

At `/playground/qubit-state`, set (α, β) via sliders or polar form and confirm normalization. Observe how changing relative phase affects X-basis probabilities while Z-basis probabilities remain fixed.

Example. The state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ yields $P(0) = P(1) = \frac{1}{2}$ upon *Z* measurement. The state $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ yields the same *Z* probabilities, yet $|+\rangle$ and $|-\rangle$ are orthogonal and physically distinguishable by *X*-basis measurement.

2.2 Complex Amplitudes

2.2.1 Complex numbers review

For $z = a + bi$ with $a, b \in \mathbb{R}$:

$$z^* = a - bi, \quad |z| = \sqrt{a^2 + b^2}, \quad \arg(z) = \tan^{-1}(b/a). \quad (2.3)$$

Polar form $z = re^{i\phi}$ uses Euler's formula $e^{i\phi} = \cos \phi + i \sin \phi$. Multiplication adds phases: $r_1 e^{i\phi_1} \cdot r_2 e^{i\phi_2} = r_1 r_2 e^{i(\phi_1 + \phi_2)}$.

2.2.2 Normalization

Given unnormalized (α, β) , divide by $\|\psi\| = \sqrt{|\alpha|^2 + |\beta|^2}$. Global phase $e^{i\gamma} |\psi\rangle$ is unobservable in standard measurements.

Interactive Lab 4: Complex Number Explorer

Explore `/playground/complex-plane`: multiply complex amplitudes graphically and verify $|z_1 z_2| = |z_1| |z_2|$ and $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$.

Example. Normalize $|\psi'\rangle = (2, 1)^\top$:

$$\|\psi'\| = \sqrt{4 + 1} = \sqrt{5}, \quad |\psi\rangle = \frac{2}{\sqrt{5}} |0\rangle + \frac{1}{\sqrt{5}} |1\rangle.$$

Then $P(0) = |\alpha|^2 = 4/5$.

2.3 Measurement in the Z Basis

2.3.1 Born rule

Measuring $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$ in the Z basis yields outcome 0 with probability $|\alpha|^2$ and outcome 1 with probability $|\beta|^2$. After outcome 0, the post-measurement state is $|0\rangle$ (up to normalization if using non-demolition models; we use the projective standard throughout).

2.3.2 Expectation value of Z

The Pauli- Z operator has eigenvalues $+1$ on $|0\rangle$ and -1 on $|1\rangle$. For pure $|\psi\rangle$,

$$\langle Z \rangle = |\alpha|^2 - |\beta|^2. \quad (2.4)$$

Interactive Lab 6: Measurement Simulator

Run `/playground/measurement` on $|\psi\rangle = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle$. Compare theoretical $P(0) = 3/4$ with histogram frequencies over many shots.

Example. For $|\psi\rangle = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle$:

$$P(0) = \frac{3}{4}, \quad P(1) = \frac{1}{4}, \quad \langle Z \rangle = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}.$$

2.4 Other Measurement Bases

2.4.1 X and Y bases

The X basis uses

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \quad (2.5)$$

The Y basis uses

$$|+i\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \quad |-i\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}. \quad (2.6)$$

Any orthonormal basis defines a measurement. Rewrite $|\psi\rangle$ in the chosen basis; squared magnitudes of coefficients give probabilities.

2.4.2 Change of basis as unitary rotation

Measuring in the X basis is equivalent to applying H (Hadamard), then measuring in Z . Similarly, $S^\dagger H$ relates Y measurements to Z .

Example. Express $|0\rangle$ in the X basis. Since $H|+\rangle = |0\rangle$ and H is unitary, $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$. Thus $P(+)=P(-)=1/2$ when measuring $|0\rangle$ in the X basis.

2.5 Global and Relative Phase

2.5.1 Global phase

Multiplying $|\psi\rangle$ by $e^{i\gamma}$ does not change any measurement statistics in any basis: probabilities depend on $|\alpha|^2$ and $|\beta|^2$ only.

2.5.2 Relative phase

The relative phase between $|0\rangle$ and $|1\rangle$ components *does* matter. Compare

$$|\psi_1\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |\psi_2\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Both have $P(0)=P(1)=1/2$ in Z , but $|\psi_1\rangle = |+\rangle$ and $|\psi_2\rangle = |-\rangle$ are orthogonal. Interference in multi-qubit algorithms depends critically on relative phases.

2.6 The Bloch Sphere

2.6.1 Parametrization

Ignoring global phase, any pure qubit state can be written

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle, \quad (2.7)$$

with Bloch coordinates

$$x = \sin \theta \cos \phi, \quad y = \sin \theta \sin \phi, \quad z = \cos \theta. \quad (2.8)$$

The surface of the unit sphere represents pure states; the interior (not realized by pure states alone) corresponds to mixed states introduced later.

2.6.2 Landmark states

$$\begin{aligned} |0\rangle &\mapsto (0, 0, 1), & |1\rangle &\mapsto (0, 0, -1), \\ |+\rangle &\mapsto (1, 0, 0), & |-\rangle &\mapsto (-1, 0, 0), \\ |+i\rangle &\mapsto (0, 1, 0), & |-i\rangle &\mapsto (0, -1, 0). \end{aligned}$$

Interactive Lab 7: 3D Bloch Sphere

Manipulate θ, ϕ at [/playground/bloch-sphere](#). Apply X, Y, Z, H and watch the state vector rotate on the sphere. Verify (2.8) numerically.

Example. The state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$ has $\theta = \pi/2$, $\phi = \pi/2$, hence Bloch vector $(0, 1, 0)$ —the north pole of the Y axis, i.e. $|+i\rangle$.

2.7 One-Qubit Gates

2.7.1 Pauli and Hadamard gates

Common gates include identity I , Pauli X, Y, Z , Hadamard H , phase S , and T :

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}. \quad (2.9)$$

Pauli gates rotate by π about their respective Bloch axes; H maps $Z \leftrightarrow X$.

2.7.2 Gate identities

$$X^2 = Y^2 = Z^2 = H^2 = I, \quad S^2 = Z, \quad T^2 = S, \quad HXH = Z, \quad HZH = X. \quad (2.10)$$

Interactive Lab 8: Quantum Gate Explorer

At [/playground/gate-explorer](#), compose sequences such as HTH and read off the effective unitary. Confirm $H|0\rangle = |+\rangle$ and $X|0\rangle = |1\rangle$.

Example. Apply H then Z to $|+\rangle$:

$$Z|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle, \quad H|-\rangle = |1\rangle.$$

The composition HZH acts as X on the Bloch sphere (up to global phase).

Exercises

1. Normalize $(1, i)^\top$ and compute $P(0)$, $P(1)$, and $\langle Z \rangle$.
2. Write $|-\rangle$ in the Z basis and compute $P(-)$ when measuring $|0\rangle$ in the X basis.
3. Parametrize $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$ on the Bloch sphere (x, y, z) .

4. Show that $\langle\psi|Z|\psi\rangle$ in (2.4) equals z from (2.8).
5. Apply S to $|+\rangle$ and give the result in the Z basis. What Bloch rotation does S implement?
6. Prove $H^2 = I$ by matrix multiplication.
7. Why does global phase $e^{i\gamma}$ not affect $\langle Z \rangle$?
8. Prepare $|+i\rangle$ using only H and S starting from $|0\rangle$.

Chapter 3

Linear Algebra for Quantum Computing

Quantum states are vectors; quantum gates are matrices. This chapter develops the Dirac *bra-ket* notation, inner products, projectors, outer products, completeness relations, and unitary matrices—the linear-algebraic language used in every subsequent chapter.

3.1 Kets and Bras

3.1.1 Column vectors and row covectors

A ket $|\psi\rangle$ denotes a column vector in $\mathbb{C}^d$. For one qubit,

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \quad \langle\psi| = [\alpha^* \quad \beta^*]. \quad (3.1)$$

The dagger (conjugate transpose) maps kets to bras: $(|\psi\rangle)^\dagger = \langle\psi|$.

3.1.2 Dual space

Bras live in the dual space—the space of linear functionals on kets. The pairing $\langle\phi|$ acting on $|\psi\rangle$ yields a complex number, defined next.

3.2 Inner Products

3.2.1 Definition and properties

For $|\phi\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$ and $|\psi\rangle = \begin{bmatrix} c \\ d \end{bmatrix}$,

$$\langle\phi|\psi\rangle = a^*c + b^*d, \quad \|\psi\| = \sqrt{\langle\psi|\psi\rangle}. \quad (3.2)$$

Properties: conjugate symmetry $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$, linearity in the second slot, and positive definiteness $\langle\psi|\psi\rangle \geq 0$ with equality iff $|\psi\rangle = 0$.

3.2.2 Orthogonality and orthonormal bases

Vectors are *orthogonal* if $\langle \phi | \psi \rangle = 0$. An *orthonormal basis* satisfies $\langle i | j \rangle = \delta_{ij}$. The computational basis (2.2) is orthonormal.

Example. Compute $\langle + | - \rangle$ using (2.5):

$$\langle + | - \rangle = \frac{1}{2}(\langle 0 | + \langle 1 |)(|0\rangle - |1\rangle) = \frac{1}{2}(1 - 1) = 0.$$

Thus $|+\rangle$ and $|-\rangle$ are orthogonal despite sharing the same Z -measurement statistics.

3.3 Projection and Measurement

3.3.1 Born rule revisited

Measuring in basis $\{|m\rangle\}$, the probability of outcome m is

$$P(m) = |\langle m | \psi \rangle|^2. \quad (3.3)$$

The amplitude $\langle m | \psi \rangle$ is sometimes called the *overlap* of $|\psi\rangle$ with $|m\rangle$.

3.3.2 Projectors

The rank-one projector onto $|m\rangle$ is

$$P_m = |m\rangle \langle m|. \quad (3.4)$$

Then $P_m^2 = P_m$, $P_m^\dagger = P_m$, and $P(m) = \langle \psi | P_m | \psi \rangle$. For non-orthogonal measurements (POVMs), generalize to positive operators summing to identity—beyond our scope here.

Example. For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$,

$$P_0 = |0\rangle \langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad P(0) = \langle \psi | P_0 | \psi \rangle = |\alpha|^2.$$

3.4 Outer Products and Completeness

3.4.1 Rank-one operators

The outer product $|\psi\rangle \langle \phi|$ is a $d \times d$ matrix. Example:

$$|1\rangle \langle 0| = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}. \quad (3.5)$$

Such operators appear in quantum channels and density matrices.

3.4.2 Resolution of the identity

Orthonormal basis $\{|i\rangle\}$ satisfies *completeness*:

$$\sum_i |i\rangle \langle i| = I. \quad (3.6)$$

In the computational basis, $|0\rangle \langle 0| + |1\rangle \langle 1| = I$. Any state expands as $|\psi\rangle = \sum_i |i\rangle \langle i|\psi\rangle$.

Example. Expand $|+\rangle$ in the Z basis:

$$|+\rangle = |0\rangle \langle 0|+\rangle + |1\rangle \langle 1|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

3.5 Unitary Matrices

3.5.1 Definition

A matrix U is *unitary* if

$$U^\dagger U = U U^\dagger = I, \quad U^{-1} = U^\dagger. \quad (3.7)$$

Unitary matrices preserve inner products: $\langle U\phi | U\psi \rangle = \langle \phi | \psi \rangle$.

3.5.2 Physical interpretation

Quantum gates are unitary evolutions (before measurement). Normalization is preserved:

$$\langle \psi' | \psi' \rangle = \langle \psi | U^\dagger U | \psi \rangle = \langle \psi | \psi \rangle. \quad (3.8)$$

Determinant has unit magnitude: $|\det U| = 1$. Eigenvalues lie on the unit circle.

Interactive Lab 9: Matrix & Unitary Checker

Enter a 2×2 matrix at </playground/unitary-checker>. Verify $H^\dagger H = I$ and confirm that a random complex matrix fails unitarity unless specially constructed.

Example. Show H is unitary by computing $H^\dagger H$:

$$H^\dagger H = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = I.$$

3.5.3 Special unitary group

Matrices with $\det U = 1$ form $SU(2)$, the double cover of rotations on the Bloch sphere. Pauli matrices X, Y, Z generate $SU(2)$ up to global phase.

3.6 Linear Operators on Composite Systems (Preview)

When subsystems combine, operators act as tensor products: $A \otimes B$ applies A to the first subsystem and B to the second. Chapter 4 develops this systematically. For now, note that unitarity extends: $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$.

Exercises

1. Compute $\langle 0|1\rangle$, $\langle +|\psi\rangle$ for $|\psi\rangle = |0\rangle$, and $\langle i|+i\rangle$ where $|i\rangle = |+i\rangle$ in the Y basis.
2. Verify (3.6) explicitly for the X basis states $|\pm\rangle$.
3. Write $|1\rangle\langle 0|$ and $|+\rangle\langle -|$ as 2×2 matrices.
4. Show that if U is unitary, so is $U^\dagger$.
5. Prove $\|U|\psi\rangle\| = \||\psi\rangle\|$ using (3.7).
6. For projector $P_+ = |+\rangle\langle +|$, compute $\text{Tr}(P_+)$ and P_+^2 .
7. Express X as a linear combination of outer products in the Z basis.
8. Why must quantum gate matrices be unitary rather than merely invertible?

Chapter 4

Composite Quantum Systems

Multiple qubits describe joint quantum systems whose state space grows exponentially: n qubits live in a 2^n -dimensional Hilbert space. This chapter introduces tensor products, entanglement, Bell states, controlled gates, the no-cloning theorem, and universal gate sets—the structural foundation for circuits, protocols, and algorithms.

4.1 Tensor Products

4.1.1 Kronecker product

For single-qubit kets $|\psi\rangle \in \mathbb{C}^2$ and $|\phi\rangle \in \mathbb{C}^2$, the *tensor product* $|\psi\rangle \otimes |\phi\rangle$ is a four-dimensional vector. Example:

$$|0\rangle \otimes |1\rangle = |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}. \quad (4.1)$$

A general two-qubit *pure* state is

$$|\Psi\rangle = c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle, \quad \sum_{i,j} |c_{ij}|^2 = 1. \quad (4.2)$$

4.1.2 Dimension counting

For n qubits, the computational basis $|b_1 \cdots b_n\rangle$ has 2^n elements. This exponential scaling is the origin of both quantum computational power and the difficulty of classical simulation.

Interactive Lab 10: Tensor Product Calculator

At `/playground/tensor-product`, form $|+\rangle \otimes |0\rangle$, $|1\rangle \otimes |-\rangle$, and verify inner products factorize on product states: $\langle\psi_1 \otimes \phi_1 | \psi_2 \otimes \phi_2\rangle = \langle\psi_1 | \psi_2\rangle \langle\phi_1 | \phi_2\rangle$.

Example. Write $|+\rangle \otimes |0\rangle$ in the four-dimensional computational basis:

$$|+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle).$$

4.2 Entanglement

4.2.1 Separable versus entangled states

A pure state $|\Psi\rangle$ on two qubits is *separable* (product) if

$$|\Psi\rangle = |\alpha\rangle \otimes |\beta\rangle \quad (4.3)$$

for some single-qubit states $|\alpha\rangle, |\beta\rangle$. Otherwise it is *entangled*. Entanglement is not correlation in the classical sense—local hidden-variable models cannot reproduce all quantum predictions (Chapter 7).

4.2.2 Bell states

The four *Bell states* form an orthonormal maximally entangled basis:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \quad |\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}, \quad (4.4)$$

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \quad |\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}. \quad (4.5)$$

Measuring one qubit of $|\Phi^+\rangle$ in Z perfectly predicts the other (00 or 11), yet individual qubits are maximally mixed.

Interactive Lab 11: Bell State Generator

Prepare each Bell state at `/playground/bell-states` using H and CNOT. Measure qubit 0 repeatedly and track correlations on qubit 1.

Example. Show $|\Phi^+\rangle$ is not separable. Suppose $|\Phi^+\rangle = (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$. Matching coefficients yields $ac = 1/\sqrt{2}$, $ad = 0$, $bc = 0$, $bd = 1/\sqrt{2}$. From $ad = 0$ and $bc = 0$, either $a = 0$ or $d = 0$; either case contradicts $ac = 1/\sqrt{2}$ or $bd = 1/\sqrt{2}$.

Interactive Lab 12: Entanglement Measurement

Use `/playground/entanglement` to estimate correlation functions for product versus Bell states. Confirm that product states factorize while Bell states violate certain classical bounds.

4.3 Multi-Qubit Gates

4.3.1 CNOT gate

The controlled-NOT (CNOT) acts on $|a, b\rangle$ as

$$\text{CNOT} |a, b\rangle = |a, a \oplus b\rangle. \quad (4.6)$$

Matrix form in the ordered basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}. \quad (4.7)$$

CNOT is entangling: $H \otimes I$ followed by CNOT maps $|00\rangle \rightarrow |\Phi^+\rangle$.

4.3.2 SWAP and Toffoli

SWAP exchanges qubits: $\text{SWAP } |a, b\rangle = |b, a\rangle$. The Toffoli (CCNOT) gate generalizes classical reversible AND to three qubits. Any classical reversible function embeds in quantum circuits using Toffoli and X gates.

Example. Apply CNOT to $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = |+\rangle \otimes |0\rangle$:

$$\text{CNOT } |+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle.$$

4.4 No-Cloning Theorem

4.4.1 Statement and proof sketch

There is no unitary U such that for all normalized $|\psi\rangle$,

$$U |\psi\rangle \otimes |0\rangle = |\psi\rangle \otimes |\psi\rangle. \quad (4.8)$$

Proof sketch. Take $\langle\phi|\psi\rangle$ for two non-orthogonal states. Unitarity requires $\langle\phi|\psi\rangle = \langle\phi|\psi\rangle^2$, forcing $\langle\phi|\psi\rangle \in \{0, 1\}$. Thus cloning works only for orthogonal states—not arbitrary unknown qubits.

4.4.2 Implications

Unknown quantum states cannot be duplicated. Error correction must encode information redundantly without copying (Chapter 5). Teleportation transfers state information without cloning (Chapter 7).

4.5 Universal Gate Sets

4.5.1 Approximate universality

A finite gate set is *universal* if any n -qubit unitary can be approximated to arbitrary precision using finitely many gates from the set. The set $\{H, T, \text{CNOT}\}$ is universal for single-qubit rotations plus entanglement.

4.5.2 Solovay–Kitaev theorem

The Solovay–Kitaev theorem (stated without proof) guarantees efficient ε -approximations of arbitrary $\text{SU}(2)$ gates using H, T and their inverses, with gate count $O(\log^c(1/\varepsilon))$ for a constant c . Combined with CNOT, this yields universal quantum computation.

Example. Approximate $R_z(\pi/8)$ using $T = \text{diag}(1, e^{i\pi/4})$ applied once. For finer angles, compose T with H and Clifford gates in Solovay–Kitaev compilations.

Exercises

1. Write $|1\rangle \otimes |+\rangle$ and $|-\rangle \otimes |-\rangle$ in the four-bit computational basis.
2. Show $|\Psi^-\rangle$ is entangled using the same coefficient-matching argument as $|\Phi^+\rangle$.
3. Compute $\text{CNOT}(|+\rangle \otimes |+\rangle)$ step by step.
4. Verify (4.7) by applying CNOT to each basis vector.
5. Explain why cloning would simplify BB84 eavesdropping but is forbidden.
6. How many parameters specify a general two-qubit pure state up to global phase? Compare with a product state.
7. Express SWAP using three CNOT gates (draw or write the sequence).
8. Why does $\{X, \text{CNOT}\}$ alone fail to be universal for arbitrary single-qubit rotations?

Chapter 5

Quantum Error Correction

Real quantum devices decohere: amplitudes leak to the environment, relative phases randomize, and energy relaxes toward ground states. *Quantum error correction* (QEC) protects *logical* qubits by encoding them redundantly across many *physical* qubits and continuously extracting error *syndromes* without learning the encoded data. This chapter covers noise models, the three-qubit bit-flip and phase-flip codes, the Shor nine-qubit code, stabilizer formalism (preview), and the distinction between mitigation and full fault tolerance.

5.1 Noise and Decoherence

5.1.1 Error channels

Common error types include:

- **Bit flip** (X): $|0\rangle \leftrightarrow |1\rangle$.
- **Phase flip** (Z): $|+\rangle \leftrightarrow |-\rangle$; $|1\rangle \mapsto -|1\rangle$.
- **Depolarizing**: random I, X, Y, Z with probabilities.
- **Amplitude damping** (T_1): relaxation toward $|0\rangle$.
- **Phase damping** (T_2): loss of off-diagonal coherence.

Open-system evolution is modeled by completely positive trace-preserving (CPTP) maps; for introductory purposes, Pauli error models suffice.

5.1.2 Hardware context (September 2026)

As of the September 2026 research cutoff, large-scale fault-tolerant quantum computation remains an active engineering goal. Experiments demonstrate improving *logical* error rates with surface-code scaling [1], while vendor roadmaps should be distinguished from established theorems.

Example. A bit-flip channel with probability p on one qubit maps $\rho \mapsto (1-p)\rho + pX\rho X$. For $p = 0.01$, repeated gates accumulate error exponentially in circuit depth unless corrected.

5.2 Three-Qubit Bit-Flip Code

5.2.1 Encoding

Logical states encode as

$$|0_L\rangle = |000\rangle, \quad |1_L\rangle = |111\rangle. \quad (5.1)$$

An arbitrary logical qubit $\alpha|0_L\rangle + \beta|1_L\rangle$ spreads information across three physical qubits.

5.2.2 Syndrome measurement

Measure parity Z_0Z_1 and Z_1Z_2 without collapsing the logical α, β (using ancilla-based circuits). Outcomes form a *syndrome* identifying which X error occurred, if any. Apply corrective X on the flagged qubit.

Interactive Lab 13: Error Correction Simulator

At `/playground/error-correction`, inject a random single bit-flip on an encoded $|+_L\rangle$ state. Read syndromes and apply correction. Verify the logical state is restored.

Example. An X error on qubit 2 flips $|010\rangle$ component in $\alpha|000\rangle + \beta|111\rangle$ if only qubit 2 errs—more generally, X_2 on $\alpha|0_L\rangle + \beta|1_L\rangle$ yields $\alpha|010\rangle + \beta|101\rangle$. Parity checks distinguish X_0, X_1, X_2 syndromes uniquely for single errors.

5.3 Phase-Flip Code

5.3.1 Encoding in X eigenbasis

Phase errors flip $|+\rangle \leftrightarrow |-\rangle$. Encode

$$|0_L\rangle = |+++ \rangle, \quad |1_L\rangle = |-- \rangle. \quad (5.2)$$

A single Z error on one qubit reverses one $|+\rangle$ to $|-\rangle$, detectable by X -parity measurements.

5.3.2 Conjugation by Hadamard

Since $HXH = Z$ and $HZH = X$, conjugating the bit-flip code circuit by $H^{\otimes 3}$ converts bit-flip protection into phase-flip protection. This *basis change* trick recurs in stabilizer codes.

5.4 Shor Nine-Qubit Code

5.4.1 Simultaneous bit- and phase-flip protection

The Shor code concatenates bit-flip and phase-flip encodings using nine physical qubits per logical qubit. It corrects arbitrary single-qubit Pauli errors. The encoding spreads one logical qubit across a block resilient to both X and Z faults on any one wire.

5.4.2 Historical role and modern successors

Shor's code established that quantum error correction is possible in principle. Modern research emphasizes *surface codes* and other topological schemes with higher thresholds and 2D-local check layouts suitable for superconducting and ion-trap arrays.

Example. Logical $|0_L\rangle$ in Shor's code is a highly entangled superposition of 512 computational basis strings (nine qubits). A single Y error (combined bit and phase) maps to a detectable syndrome without revealing whether the logical state was $|0_L\rangle$ or $|1_L\rangle$.

5.5 Stabilizer Formalism (Preview)

5.5.1 Stabilizer group

Many QEC codes are described by an abelian *stabilizer group* $\mathcal{S}$ of Pauli operators that fix the code space:

$$|\psi_L\rangle \in \mathcal{C} \quad \Leftrightarrow \quad s|\psi_L\rangle = |\psi_L\rangle \quad \forall s \in \mathcal{S}. \quad (5.3)$$

Measuring stabilizer generators yields syndromes without collapsing logical information when performed via ancilla coupling. The three-qubit bit-flip code has stabilizers Z_0Z_1 and Z_1Z_2 ; the phase-flip code uses X_0X_1 and X_1X_2 .

5.5.2 CSS codes

Calderbank–Shor–Steane (CSS) codes separate X -type and Z -type stabilizers, enabling bitwise error correction for each type. The Steane $[[7,1,3]]$ code uses seven physical qubits per logical qubit and corrects any single Pauli error—a stepping stone between three-qubit toy codes and surface codes.

Example. For the bit-flip code, valid codewords satisfy $Z_0Z_1|\psi\rangle = |\psi\rangle$ and $Z_1Z_2|\psi\rangle = |\psi\rangle$. An X error on qubit 1 anticommutes with Z_0Z_1 , flipping the syndrome bit associated with that stabilizer.

5.5.3 Surface code glance

The surface code stabilizes a 2D lattice of physical qubits with local X - and Z -type checks on plaquettes and stars. Logical operators correspond to non-contractible loops on the surface. Threshold estimates near 1% physical error per gate motivate current hardware efforts [1]. Distance d codes correct up to $\lfloor (d-1)/2 \rfloor$ Pauli errors.

5.5.4 Fault-tolerant gate synthesis

Logical gates are implemented by *transversal* operations (applying physical gates in parallel without spreading errors) or *code switching* and *magic-state injection* for non-Clifford gates. The full pipeline—encode, syndrome extract, decode, distill—defines the resource overhead separating NISQ demos from cryptographically relevant computation.

5.6 Error Mitigation versus Correction

5.6.1 NISQ-era techniques

In the NISQ (noisy intermediate-scale quantum) era [7], *error mitigation* reduces bias without full encoding:

- **Zero-noise extrapolation:** run at multiple noise scales and extrapolate to zero.
- **Probabilistic error cancellation:** combine noisy circuits with signed weights.
- **Readout error mitigation:** calibrate and invert measurement confusion matrices.

5.6.2 When mitigation suffices

Mitigation helps variational algorithms and short-depth demonstrations. It is *not* a substitute for fault tolerance at scale: deep circuits amplify errors exponentially without logical encoding and continual syndrome extraction.

Example. Suppose a depolarizing error probability $p = 10^{-3}$ per gate and depth $D = 1000$. Naively, success probability $\approx (1 - p)^D \approx e^{-1} \approx 0.37$. Mitigation might improve effective fidelity modestly; fault-tolerant QEC targets logical error rates $\ll p$ per logical gate.

Exercises

1. What syndrome pattern indicates an X error on qubit 2 in the three-bit bit-flip code?
2. Why cannot you clone $|\psi\rangle$ to protect against erasure?
3. Distinguish physical and logical qubits in one paragraph.
4. Write stabilizer generators for the three-qubit phase-flip code.
5. If two independent bit-flips occur, can the three-qubit code always correct? Explain.
6. Compare resource overhead: three-qubit code versus Shor nine-qubit code per logical qubit.
7. Name one mitigation technique and state its limitation relative to fault tolerance.
8. Why must syndrome measurements be designed not to reveal α, β in $\alpha|0_L\rangle + \beta|1_L\rangle$?

Chapter 6

Quantum Circuits and Programming

Quantum algorithms are expressed as *circuits*: wires represent qubits, time flows left to right, boxes are gates, control dots mark controlled operations, and measurement symbols denote readout. This chapter formalizes the circuit model, walks through Bell-state synthesis, surveys gate sets and OpenQASM, and discusses depth and resource costs—connecting mathematics to the software tools used on real hardware.

6.1 The Circuit Model

6.1.1 Unitary evolution

An n -qubit circuit transforms $|\psi_{\text{in}}\rangle$ to

$$|\psi_{\text{out}}\rangle = U_k \cdots U_2 U_1 |\psi_{\text{in}}\rangle, \quad (6.1)$$

where each U_i acts unitarily on some subset of qubits (often tensored with identity on others). Measurement at the end samples classical outcomes from Born probabilities. Circuits are *acyclic*: no feedback unless classical control is explicitly modeled.

6.1.2 Diagrammatic conventions

Wires index qubits from top to bottom; time flows left to right. A solid control dot on one wire with $\oplus$ on another denotes CNOT. Single-qubit boxes label gates (H , T , etc.). Double lines occasionally denote classical bits conditioned on measurement outcomes.

6.1.3 Depth, width, and gate count

- **Width:** number of qubits n .
- **Gate count:** total gates applied.
- **Depth:** number of sequential *layers* of gates that can run in parallel on disjoint qubits.

Depth determines latency on hardware with sequential gate calibration; gate count relates to total error budget. On superconducting devices, two-qubit gates along the same coupling edge often cannot execute simultaneously.

Example. *Two CNOTs sharing a control qubit cannot run in parallel if they use the same control line in conflicting ways; two single-qubit gates on different qubits can occupy one depth layer. A circuit with n sequential CNOTs on a chain has depth $\Omega(n)$ even if gate count is also n .*

6.1.4 Measurement and classical control

Mid-circuit measurement feeds classical registers that condition later gates (*dynamic circuits*). This extends (6.1) to conditional operations $U_k(c_1, \dots, c_m)$ where c_j are prior measurement outcomes—essential for teleportation and error correction.

6.2 Building Bell States

6.2.1 Standard construction

The canonical Bell circuit prepares $|\Phi^+\rangle$ from $|00\rangle$:

$$|00\rangle \xrightarrow{H_0} \xrightarrow{\text{CNOT}_{0,1}} |\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \quad (6.2)$$

Here H_0 denotes Hadamard on qubit 0; $\text{CNOT}_{0,1}$ uses qubit 0 as control and qubit 1 as target.

6.2.2 Step-by-step state update

$$\begin{aligned} |00\rangle &\xrightarrow{H \otimes I} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle), \\ &\xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle. \end{aligned}$$

Other Bell states arise from additional single-qubit Pauli rotations before or after this core.

Interactive Lab CB: Circuit Builder

Open `/playground/circuit-builder`. Drag H and CNOT onto two qubits, simulate step-by-step, and read the statevector. Confirm amplitudes match (4.5).

6.3 Gate Sets and Native Hardware

6.3.1 Universal versus native gates

Theory uses $\{H, T, \text{CNOT}\}$; hardware exposes *native* gates—e.g. R_x, R_y, R_z and CNOT on fixed connectivity graphs. *Compilation* transpiles abstract circuits into native pulses with SWAP routing for limited connectivity.

6.3.2 Clifford+T and T-count

Fault-tolerant architectures often separate fast *Clifford* gates (H, S, CNOT) from costly T gates, which require magic-state distillation. T -count dominates resource estimates for Shor’s algorithm.

6.4 OpenQASM

6.4.1 Language overview

Open Quantum Assembly Language (OpenQASM) describes circuits textually. OpenQASM 3.0 adds classical control flow, gate modifiers, and richer typing. A minimal Bell preparation:

```

OPENQASM 3.0;
include "stdgates.inc";
qubit[2] q;
h q[0];
cx q[0], q[1];

```

6.4.2 Registers and timing

OpenQASM declares `qubit[n]` quantum and `bit[m]` classical registers. Statements execute in order unless grouped into `box` blocks for timing semantics. Gate arguments specify qubit indices; `creg` stores measurement outcomes for conditional gates such as `if (c == 1) x q[0];`.

6.4.3 Reading and writing QASM

Export from visual builders or SDKs (Qiskit `qasm3.dumps`, Cirq `to_qasm`). Always map each statement back to unitaries on the Hilbert space; `cx` is CNOT, `h` is Hadamard, `rz(theta)` is $R_z(\theta)$.

Example. *The OpenQASM snippet above corresponds to $U = \text{CNOT}_{0,1}(H \otimes I)$ applied left-to-right as $U|00\rangle = |\Phi^+\rangle$. Reversing gate order in the file changes the unitary product accordingly.*

Example. *The unitary for $H \otimes I$ on two qubits is*

$$H \otimes I = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

Apply it to $|00\rangle$ before CNOT to derive (6.2).

6.5 Depth and Resource Analysis

6.5.1 Ripple adders revisited

An n -bit ripple-carry adder built from Toffoli and CNOT has depth $O(n)$ in the worst case—analogue classical depth from Chapter 1. Quantum carry-lookahead and Fourier adders reduce depth at the cost of ancillas.

6.5.2 Simulation cost

Exact state-vector simulation of n qubits requires $O(2^n)$ complex amplitudes. Clifford circuits (generated by $\{H, S, \text{CNOT}\}$ without T) simulate in polynomial time (Gottesman–Knill), explaining why random-circuit supremacy demos use non-Clifford gates [2].

6.6 Modern Software Ecosystem (2026)

Production quantum software typically combines:

- **Circuit construction:** OpenQASM 3, Qiskit, Cirq, PennyLane, and vendor SDKs.
- **Simulation:** local state-vector simulators (exponential in n) and stabilizer simulators for Clifford circuits.

- **Execution:** cloud access to superconducting, trapped-ion, neutral-atom, and photonic backends with calibration data and error models.

Map API calls (`h(0)`, `cx(0,1)`) to unitaries and tensor structure before trusting black-box outputs.

Exercises

1. Write the 4×4 matrix for $H \otimes I$ and verify $(H \otimes I) |00\rangle$.
2. Count depth (qualitatively) of an n -bit ripple-carry adder from Toffoli gates.
3. Export a Bell circuit to OpenQASM: which gates appear?
4. Why does CNOT not equal $\text{CNOT}_{1,0}$? Give the matrix of swapped control/target.
5. Estimate memory for simulating $n = 30$ qubits in double precision (bytes $\approx 16 \cdot 2^n$).
6. Which gates in $\{H, S, \text{CNOT}\}$ generate the Clifford group?
7. Transpile T into $R_z(\pi/4)$ up to global phase.
8. Explain why measurement is not modeled as a unitary gate in (6.1).

Chapter 7

Entanglement and Quantum Protocols

Entanglement enables communication and cryptography protocols impossible with classical bits alone, while respecting the *no-signaling* principle: local measurements cannot transmit controllable faster-than-light messages. This chapter covers EPR paradox and hidden variables, the CHSH inequality, superdense coding, quantum teleportation, and BB84 quantum key distribution.

7.1 EPR and Hidden Variables

7.1.1 The EPR argument

Einstein, Podolsky, and Rosen (1935) highlighted that entangled pairs exhibit perfect correlations when both parties measure in the same basis, yet each qubit alone appears maximally random. They questioned whether quantum mechanics is *complete* or whether *local hidden variables* (LHV) predetermine outcomes.

7.1.2 Local realism versus quantum predictions

An LHV model assigns definite values to observables before measurement, with outcomes determined locally without faster-than-light influence. Bell and CHSH showed certain correlations violate inequalities any LHV model must satisfy.

Example. *Shared $|\Phi^+\rangle$: Alice and Bob both measure Z . Outcomes are perfectly correlated (00 or 11) yet individually random with $P(0) = P(1) = 1/2$. Classically, one might imagine hidden bits copied at source; Bell tests rule out local hidden variables for maximal entanglement.*

7.2 CHSH Inequality

7.2.1 Setup

Alice chooses measurement settings $a, a' \in \{0, 1\}$; Bob chooses b, b' . Define correlation

$$E(a, b) = P(00) + P(11) - P(01) - P(10) \tag{7.1}$$

for ± 1 outcomes mapped to $\{+1, -1\}$.

7.2.2 Classical and quantum bounds

The CHSH combination

$$S = E(a, b) + E(a, b') + E(a', b) - E(a', b') \quad (7.2)$$

satisfies $|S| \leq 2$ for any local hidden-variable theory. Quantum mechanics permits $|S| \leq 2\sqrt{2}$ (Tsirelson bound), achieved by suitable measurements on $|\Phi^-\rangle$ or singlet states.

Interactive Lab 14: Bell/CHSH Experiment

Simulate `/playground/chsh`: choose measurement angles, estimate $E(a, b)$ from shot statistics, and compute S . Push toward the quantum limit $2\sqrt{2}$.

Example. For singlet $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$, opposite measurements in the Bloch plane can yield $E = -\cos(\theta_A - \theta_B)$. Optimizing settings gives $|S| = 2\sqrt{2}$.

7.3 Superdense Coding

7.3.1 Protocol

Alice and Bob share $|\Phi^+\rangle_{AB}$. Alice encodes two classical bits into one qubit by applying one of $\{I, X, Z, XZ\}$ to her half, then sends her qubit to Bob. Bob performs a Bell-basis measurement decoding $(00), (01), (10), (11)$.

7.3.2 Resource trade-off

Two classical bits travel on one qubit *plus* one shared ebit (entang bit). Entanglement is consumable but not cloneable; pre-shared entanglement is a resource.

Interactive Lab 15: Superdense Coding

Walk through `/playground/superdense`: pick a two-bit message, apply Alice's encoding, and verify Bob's Bell measurement recovers the message with certainty.

Example. Alice sends bit pair $(1, 0)$ using Z . On $|\Phi^+\rangle$, $Z_A |\Phi^+\rangle = |\Phi^-\rangle$. Bob's Bell measurement distinguishes $|\Phi^+\rangle$ from $|\Phi^-\rangle$, decoding $(1, 0)$.

7.4 Quantum Teleportation

7.4.1 Protocol steps

Unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ on qubit 0 teleports to Bob using shared $|\Phi^+\rangle_{12}$ on qubits 1 (Alice) and 2 (Bob):

1. Alice applies $\text{CNOT}_{0 \rightarrow 1}$ then H_0 .
2. Alice measures qubits 0 and 1 in Z , obtaining two classical bits.
3. Bob applies corrections $X^{m_1} Z^{m_0}$ to qubit 2 based on outcomes (m_0, m_1) .

Bob's qubit ends in $|\psi\rangle$; Alice's original is destroyed.

7.4.2 No cloning and no signaling

Teleportation transfers quantum state using one ebit and two classical bits. It does not copy $|\psi\rangle$ (no-cloning) and cannot signal faster than light because Bob's state is mixed until classical bits arrive.

Interactive Lab 16: Teleportation Simulator

Run `/playground/teleportation` with random $|\psi\rangle$. Track Alice's measurements and Bob's conditional corrections.

Example. After Alice measures $|\Phi^+\rangle$ shared pair and gets $(m_0, m_1) = (0, 0)$, Bob's qubit is $X^0 Z^0 |\psi\rangle = |\psi\rangle$ without correction if the circuit normalization matches standard conventions; other outcomes require X and/or Z .

7.5 BB84 Quantum Key Distribution

7.5.1 Protocol outline

1. Alice chooses random bits and random bases (Z or X), encodes, and sends qubits.
2. Bob measures in random bases.
3. They publicly compare bases (not outcomes); matching bases yield sifted key bits.
4. They sample a subset to estimate error rate; elevated errors suggest eavesdropping.

7.5.2 Security intuition

An eavesdropper Eve cannot measure unknown states without disturbance (no-cloning, measurement back-action). Intercept-resend attacks increase bit-error rates detectable in the sifted key.

Interactive Lab 17: BB84 Simulator

Use `/playground/bb84` with and without an eavesdropper. Compare error rates on the sifted key.

Example. Alice encodes bit 1 in X basis as $|-\rangle$. Bob measures in Z , obtaining random outcomes—sifted key uses only rounds where bases matched. If Eve intercepts with random bases, $\approx 25\%$ of sifted bits err for unbiased encoding.

Exercises

1. After Alice measures her qubit of $|\Phi^+\rangle$ in Z and gets 1, what is Bob's qubit (before any correction)?
2. Why does Eve's intercept-resend attack increase the error rate in BB84?
3. Sketch how S in (7.2) is computed from four correlation settings.
4. How many classical bits and ebits does teleportation consume per unknown qubit?
5. Can superdense coding work without pre-shared entanglement? Justify briefly.
6. Write the four Bell measurement projectors as 4×4 matrices.

7. Explain no-signaling: why can't Alice's local choice of Z versus X measurement send a message to Bob instantly?
8. Compare BB84 security assumptions with RSA security assumptions in one paragraph.

Chapter 8

Quantum Algorithms

Quantum algorithms exploit interference and phase to outperform classical strategies for *specific* problems. This chapter develops the oracle model, Deutsch and Deutsch–Jozsa algorithms, Bernstein–Vazirani, Simon’s algorithm, Grover search, the quantum Fourier transform, phase estimation, and Shor’s factoring algorithm—with complexity comparisons throughout.

8.1 Oracles and Phase Kickback

8.1.1 Standard oracle

A Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is accessed via a unitary

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle. \quad (8.1)$$

The second register is an *ancilla*; XOR implements reversible function evaluation.

8.1.2 Phase oracle via kickback

Initialize the ancilla in $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Then

$$U_f |x\rangle |-\rangle = |x\rangle \frac{(-1)^{f(x)} |0\rangle - |1\rangle}{\sqrt{2}} = (-1)^{f(x)} |x\rangle |-\rangle. \quad (8.2)$$

Ignoring the unchanged ancilla defines the *phase oracle*

$$O_f |x\rangle = (-1)^{f(x)} |x\rangle. \quad (8.3)$$

Phase kickback is the workhorse of many algorithms.

Interactive Lab 18: Deutsch Algorithm

Implement `/playground/deutsch`: test constant versus balanced $f : \{0, 1\} \rightarrow \{0, 1\}$ in one query using phase kickback.

Example. For $f(x) = x$ (balanced), $O_f |0\rangle = +|0\rangle$ and $O_f |1\rangle = -|1\rangle$: the oracle acts as Z on the query qubit—detectable after H interference.

8.2 Deutsch and Deutsch–Jozsa

8.2.1 Deutsch’s problem

For $n = 1$, distinguish *constant* ($f(0) = f(1)$) from *balanced* ($f(0) \neq f(1)$) functions. Classically, two queries may be needed in the worst case; Deutsch’s algorithm uses one.

8.2.2 Circuit outline

Prepare $|+\rangle|-\rangle$, apply O_f , apply H on the query qubit, measure. Constant f yields $|0\rangle$ with certainty; balanced f yields $|1\rangle$.

8.2.3 Deutsch–Jozsa generalization

For $f : \{0, 1\}^n \rightarrow \{0, 1\}$ promised constant or balanced, classical worst-case queries are $2^{n-1} + 1$; Deutsch–Jozsa uses one query with n qubits plus ancilla.

Interactive Lab 19: Deutsch–Jozsa

Explore [/playground/deutsch-jozsa](#) for $n = 2, 3$. Verify constant functions never produce all-ones parity pattern on the query register.

Example. The four $n = 1$ functions: $f_0(x) = 0$, $f_1(x) = 1$ (*constant*); $f_2(x) = x$, $f_3(x) = 1 - x$ (*balanced*). Deutsch distinguishes the two classes in one shot.

8.3 Bernstein–Vazirani

8.3.1 Hidden string problem

Given $f(x) = s \cdot x \bmod 2$ (inner product mod 2) for hidden $s \in \{0, 1\}^n$, find s . Classically, n queries are required (one per bit); Bernstein–Vazirani uses one query after preparing $|+\rangle^{\otimes n}$ and applying O_f .

8.3.2 Query complexity

The algorithm demonstrates an exponential separation in query complexity for this structured promise problem—a template for Simon and Shor.

Interactive Lab 20: Bernstein–Vazirani

At [/playground/bernstein-vazirani](#), choose random s and recover it from measurement outcomes in the Z basis after $H^{\otimes n}$.

8.4 Simon’s Algorithm

8.4.1 Periodicity promise

For $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$ with hidden period $s \neq 0$ such that $f(x) = f(x \oplus s)$, find s . Classical algorithms need exponential queries; Simon needs $O(n)$ quantum queries plus classical post-processing.

8.4.2 Linear algebra over $\mathbb{F}_2$

Measurement outcomes y satisfy $y \cdot s = 0 \pmod{2}$. Collecting $\approx n$ independent equations yields s by Gaussian elimination over $\mathbb{F}_2$.

Interactive Lab 21: Simon's Algorithm

Simulate `/playground/simon`: observe random y strings orthogonal to s and solve for s classically.

Example. If $s = 110$ and measured $y_1 = 101$, $y_2 = 011$, then $y_1 \cdot s = 1 + 0 + 0 = 1 \neq 0$ —discard dependent samples until independent constraints determine s .

8.5 Grover Search

8.5.1 Unstructured search

Search space size $N = 2^n$; exactly one marked state $|w\rangle$. Classically, $\Theta(N)$ queries are needed in the worst case; Grover achieves $O(\sqrt{N})$ oracle queries [4].

8.5.2 Grover iterate

Define oracle $O_w = I - 2|w\rangle\langle w|$ and diffusion operator $D = 2|s\rangle\langle s| - I$ where $|s\rangle = \frac{1}{\sqrt{N}} \sum_x |x\rangle$. The iterate $G = DO_w$ rotates the amplitude vector toward $|w\rangle$ by angle $\approx 2 \arcsin(1/\sqrt{N})$ per application.

8.5.3 Optimal iteration count

Approximately

$$k \approx \left\lfloor \frac{\pi}{4} \sqrt{N} \right\rfloor \quad (8.4)$$

iterations maximize success probability.

Interactive Lab 22: Grover Search

Run `/playground/grover` with $N = 8$, marked $|011\rangle$. Compare $k = 2$ versus $k = 5$ success rates with (8.4).

Example. For $N = 8$, $\lfloor \frac{\pi}{4} \sqrt{8} \rfloor = \lfloor 2.22 \rfloor = 2$. After two Grover iterations starting from uniform superposition, $P(|w\rangle) \approx 1$ for the unique marked state.

8.6 Quantum Fourier Transform

8.6.1 Definition

On $N = 2^n$ basis states,

$$\text{QFT}_N |x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i x k / N} |k\rangle. \quad (8.5)$$

The QFT is unitary and efficiently implementable with $O(n^2)$ gates, but amplitudes are *not* all readable at once—measurement samples one outcome.

Interactive Lab 23: QFT Visualizer

Inspect `/playground/qft` for $n = 3$: apply QFT to $|001\rangle$ and compare magnitude patterns before measurement.

8.7 Quantum Phase Estimation

8.7.1 Problem

Given unitary U and eigenstate $|u\rangle$ with $U|u\rangle = e^{2\pi i\phi}|u\rangle$, estimate ϕ to m bits using m ancilla qubits and controlled powers U^{2^j} .

8.7.2 Binary fraction expansion

Phase estimation implements the QFT on controlled- U outcomes, yielding an m -bit approximation to ϕ . Shor's algorithm uses QPE on modular exponentiation unitaries.

Interactive Lab 24: Phase Estimation

Use `/playground/phase-estimation` with $U = R_z$ on an eigenstate and $m = 3$. Compare readout to $\phi = 1/3 \approx 0.333\dots$ in binary.

Example. With $m = 3$, binary digits approximate $\phi = 1/3 = 0.\overline{010}_2$. Rounding errors arise from finite m ; repeat with larger m for precision.

8.8 Shor's Factoring Algorithm

8.8.1 Reduction to period finding

Factoring N reduces to finding period r of $f(x) = a^x \bmod N$ for random a coprime to N . If r is even and $a^{r/2} \not\equiv -1 \pmod{N}$, then $\gcd(a^{r/2} \pm 1, N)$ yields nontrivial factors.

8.8.2 Example $N = 15$, $a = 2$

The period is $r = 4$ since $2^4 = 16 \equiv 1 \pmod{15}$. Then

$$\gcd(2^2 + 1, 15) = \gcd(5, 15) = 5, \quad \gcd(2^2 - 1, 15) = \gcd(3, 15) = 3.$$

Quantum speedup lies in finding r via QPE; remaining steps are classical [8].

Interactive Lab 25: Period Explorer

Explore `/playground/period-finding` for modular exponentiation and period r of $f(x) = a^x \bmod N$.

Interactive Lab 26: Shor Demo

Complete the factoring pipeline at `/playground/shor` using QPE-derived periods and classical gcd steps.

8.8.3 Complexity summary

Table 8.1 collects query/time scalings covered in this chapter; see Appendix C for a fuller reference.

Table 8.1: Representative query complexity advantages (promise problems).

Problem	Classical (worst case)	Quantum
Deutsch ($n = 1$)	2 queries	1
Deutsch–Jozsa	$2^{n-1} + 1$	1
Bernstein–Vazirani	n	1
Simon	$O(2^{n/2})$	$O(n)$
Grover	$O(N)$	$O(\sqrt{N})$
Factoring (Shor)	no known poly-time	poly-time (ideal)

Exercises

1. Which of the four $n = 1$ Boolean functions are constant versus balanced?
2. Run Grover on $N = 8$ with marked $|011\rangle$; how many iterations does (8.4) predict?
3. For QPE with $m = 3$, what binary fraction approximates $\phi = 1/3$?
4. Construct a phase oracle for $f(x) = 1$ (constant) and show Deutsch outputs $|0\rangle$.
5. Why does Grover not achieve exponential speedup for NP-complete search?
6. Write the QFT matrix for $N = 4$ and verify unitarity.
7. For Simon, why are measured y vectors uniform over $\{0, 1\}^n$ before post-processing?
8. Outline Shor’s classical post-processing after obtaining period r .

Chapter 9

Modern Quantum Computing in 2026

This chapter surveys the landscape a serious introductory student should know beyond foundational algorithms: hardware platforms, surface codes and logical qubits, the NISQ era, software ecosystems, fault-tolerance roadmaps, and post-quantum cryptography. Statements are tagged by evidence type; company roadmaps are not scientific facts and must be verified against peer-reviewed literature.

9.1 Physical Qubit Platforms

9.1.1 Leading modalities

As of the September 2026 cutoff, several qubit modalities compete:

- **Superconducting transmons:** microwave control, fast gates, planar fabrication; short T_1, T_2 relative to ions.
- **Trapped ions:** long coherence, high fidelity, all-to-all connectivity in chains; slower gates.
- **Neutral atoms:** reconfigurable arrays via optical tweezers; promising for error-correction layouts.
- **Photonic:** room-temperature operation; challenges in deterministic two-qubit gates.
- **Spin qubits:** semiconductor integration potential; scaling research active.

9.1.2 Metrics that matter

Compare platforms using gate fidelity ($> 99.9\%$ two-qubit targets), coherence times (T_1, T_2), connectivity graph, and reproducible cross-calibrated benchmarks—not qubit count alone.

Example. *A device advertising “1000+ qubits” with 10^{-2} two-qubit error rates cannot yet run fault-tolerant Shor at cryptographically relevant scales without extensive QEC overhead [7].*

9.2 Logical Qubits and the Surface Code

9.2.1 From physical to logical

Fault-tolerant computation stores information in *logical* qubits encoded across many *physical* qubits. Syndrome extraction detects errors faster than they accumulate. The *threshold theorem* guarantees arbitrary low logical error rates if physical errors per gate stay below a code-specific threshold.

9.2.2 Surface code highlights

The surface code [1] arranges stabilizer checks on a 2D lattice with local interactions suitable for superconducting and neutral-atom layouts. Recent experiments demonstrate *logical* error rates below *physical* rates at increasing code distance—a milestone toward scalable QEC, though resource estimates for chemistry or RSA-scale factoring remain substantial.

Example. *Distance- d surface code requires $\approx d^2$ physical qubits per logical qubit and d rounds of syndrome extraction per logical gate in rough estimates; exact constants depend on implementation and magic-state factories.*

9.3 Noise, Channels, and Mitigation

9.3.1 NISQ definition

NISQ devices [7] have 50–500 noisy qubits without full error correction. Useful tasks include variational algorithms, simulation prototypes, and benchmarking—not arbitrary-depth fault-tolerant programs.

9.3.2 Mitigation versus correction

Mitigation (Chapter 5) reduces statistical bias in short circuits; correction suppresses logical error rates exponentially given sufficient redundancy. Long-term utility requires the latter for deep algorithms.

9.4 Quantum Advantage and Benchmarks

9.4.1 Random circuit sampling

Random circuit sampling milestones [2] demonstrate quantum behavior difficult to simulate classically for *specific* tasks. This is not universal speedup for everyday workloads.

9.4.2 How to read claims

Always ask: *Which problem? Which metric? Which classical comparison?* Cross-entropy benchmarks, quantum volume, and application-specific demos measure different things.

Example. *Sycamore-style supremacy argues classical simulation of a particular random circuit distribution is infeasible; it does not imply Gmail runs faster on quantum processors.*

9.5 Quantum Networking and Cryptography

9.5.1 Quantum key distribution

QKD (Chapter 7) offers information-theoretic security under explicit device and channel trust models. Deployment exists in limited links; scaling requires quantum repeaters still under research.

9.5.2 Post-quantum classical cryptography

Shor’s algorithm threatens RSA and elliptic-curve cryptography by efficiently solving hidden subgroup problems underlying those schemes. *Post-quantum cryptography* (PQC) standardizes classical algorithms believed secure against quantum adversaries (lattice-based, hash-based, code-based). PQC deployment proceeds independently of quantum hardware timelines.

9.6 Software Ecosystem (2026)

9.6.1 Stack layers

- **Languages & IR:** OpenQASM 3, Quil, vendor DSLs.
- **SDKs:** Qiskit, Cirq, PennyLane, Braket, Azure Quantum, IBM Quantum Platform [5].
- **Compilers:** routing, pulse-level optimization, error-aware transpilation.
- **Simulators:** state-vector, density matrix, stabilizer, tensor-network.

9.6.2 Hybrid workflows

Near-term applications combine classical optimizers with quantum coprocessors (variational quantum eigensolvers, QAOA). HPC integration treats quantum devices as specialized accelerators with queueing and calibration schedules.

9.7 Fault Tolerance Roadmaps

9.7.1 Magic states and distillation

Non-Clifford gates (especially T) dominate resource estimates. *Magic-state distillation* factories supply clean T states to fault-tolerant circuits; their footprint often exceeds the computation proper.

9.7.2 Company versus science

Vendor announcements (e.g. processor codenames, qubit counts) are engineering milestones, not proofs of utility. Distinguish peer-reviewed logical-qubit demonstrations [1] from marketing materials [3].

9.8 Resource Estimation and Classical Simulation

Exact simulation of n qubits needs $O(2^n)$ complex amplitudes—infeasible beyond ≈ 50 qubits on classical supercomputers for generic states. Clifford circuits simulate in polynomial time, clarifying why non-Clifford content matters for advantage claims.

9.9 Open Questions at the September 2026 Cutoff

- Fault-tolerant logical qubits at scale with practical space-time overheads.
- Durable quantum advantage in chemistry, optimization, or machine learning.
- Integration of quantum accelerators with classical HPC and secure PQC migration.
- Standardized benchmarks correlating device metrics with application performance.

Further Reading

See [6, 7, 5] and peer-reviewed surveys. Re-verify hardware claims against primary literature before citing in research.

Exercises

1. Compare two hardware platforms on fidelity, coherence, and connectivity in a table.
2. Explain why surface codes favor 2D-local architectures.
3. Distinguish quantum advantage, quantum supremacy, and fault tolerance in one paragraph each.
4. Why does PQC migration not require a working quantum computer?
5. Estimate qubits needed to factor RSA-2048 using rough literature exponents (order-of-magnitude).
6. Name three software layers in the 2026 quantum stack.
7. What role do magic states play in fault-tolerant T gates?
8. Critique a hypothetical headline: “1000-qubit computer breaks encryption.”

Appendix A

Gate Reference

This appendix collects matrix definitions, action rules, and composition identities for gates used throughout the text.

A.1 Single-Qubit Gates

A.1.1 Pauli and Clifford gates

Standard single-qubit gates:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad (\text{A.1})$$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}, \quad S^\dagger = \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix}, \quad T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}. \quad (\text{A.2})$$

Action on computational basis:

$$\begin{array}{ll} X|0\rangle = |1\rangle, & X|1\rangle = |0\rangle, \\ Z|0\rangle = |0\rangle, & Z|1\rangle = -|1\rangle, \\ H|0\rangle = |+\rangle, & H|1\rangle = |-\rangle. \end{array}$$

A.1.2 Rotation gates

Euler rotations about Bloch axes:

$$R_x(\theta) = e^{-i\theta X/2} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} X, \quad (\text{A.3})$$

$$R_y(\theta) = e^{-i\theta Y/2} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} Y, \quad (\text{A.4})$$

$$R_z(\theta) = e^{-i\theta Z/2} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} Z. \quad (\text{A.5})$$

Note $T = R_z(\pi/4)$ up to global phase.

A.2 Two- and Three-Qubit Gates

A.2.1 CNOT and variants

Controlled-NOT in ordering $|00\rangle, |01\rangle, |10\rangle, |11\rangle$:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \text{CNOT} |a, b\rangle = |a, a \oplus b\rangle. \quad (\text{A.6})$$

Controlled-Z (CZ) differs by local phases; $\text{CZ} = (I \otimes H) \text{CNOT} (I \otimes H)$ up to conventions.

A.2.2 SWAP and Toffoli

SWAP exchanges qubits. Toffoli (CCNOT) flips target when both controls are $|1\rangle$:

$$\text{CCNOT} |a, b, c\rangle = |a, b, c \oplus ab\rangle. \quad (\text{A.7})$$

SWAP on qubits 1,2 decomposes as three CNOTs: $\text{CNOT}_{1,2} \text{CNOT}_{2,1} \text{CNOT}_{1,2}$.

A.3 Useful Identities

A.3.1 Conjugation and commutation

$$H X H = Z, \quad H Z H = X, \quad H^2 = I, \quad (\text{A.8})$$

$$S^2 = Z, \quad T^2 = S, \quad TS = i ST \text{ (up to global phase conventions)}. \quad (\text{A.9})$$

A.3.2 Pauli algebra

$$\sigma_i \sigma_j = \delta_{ij} I + i \sum_k \epsilon_{ijk} \sigma_k, \quad \{\sigma_i, \sigma_j\} = 2\delta_{ij} I, \quad (\text{A.10})$$

for $\sigma_i \in \{X, Y, Z\}$ and Levi-Civita symbol ϵ_{ijk} .

A.4 Controlled-U Pattern

For single-qubit U , controlled- U acts as U on target when control is $|1\rangle$:

$$\text{CU} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U. \quad (\text{A.11})$$

Phase kickback uses CU with target initialized to $|-\rangle$.

Appendix B

Formula Sheet

This appendix summarizes key formulas for exam review. Cross-references point to defining chapters.

B.1 Single-Qubit States and Measurement

Pure state (Chapter 2):

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (\text{B.1})$$

Born rule (Chapter 3):

$$P(m) = |\langle m|\psi\rangle|^2, \quad \langle A \rangle = \langle \psi| A |\psi\rangle. \quad (\text{B.2})$$

Bloch parametrization:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle, \quad (x, y, z) = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta). \quad (\text{B.3})$$

B.2 Linear Algebra

Inner product and norm:

$$\langle \phi|\psi\rangle = \sum_i a_i^* b_i, \quad \|\psi\| = \sqrt{\langle \psi|\psi\rangle}. \quad (\text{B.4})$$

Projector and completeness:

$$P_m = |m\rangle \langle m|, \quad \sum_m P_m = I. \quad (\text{B.5})$$

Unitary evolution:

$$U^\dagger U = I, \quad |\psi'\rangle = U |\psi\rangle. \quad (\text{B.6})$$

B.3 Multi-Qubit Systems

Tensor product dimension: n qubits $\Rightarrow$ dimension 2^n .

Bell states (Chapter 4):

$$|\Phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \quad |\Psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}. \quad (\text{B.7})$$

CNOT: $\text{CNOT} |a, b\rangle = |a, a \oplus b\rangle$.

B.4 Error Correction

Three-qubit bit-flip encoding:

$$|0_L\rangle = |000\rangle, \quad |1_L\rangle = |111\rangle. \quad (\text{B.8})$$

Stabilizer example: Z_0Z_1 and Z_1Z_2 for bit-flip code.

B.5 Oracles and Algorithms

Phase oracle from kickback:

$$O_f |x\rangle = (-1)^{f(x)} |x\rangle. \quad (\text{B.9})$$

Grover operators:

$$O_w = I - 2|w\rangle\langle w|, \quad D = 2|s\rangle\langle s| - I, \quad k \approx \left\lfloor \frac{\pi}{4} \sqrt{N} \right\rfloor. \quad (\text{B.10})$$

Quantum Fourier transform:

$$\text{QFT}_N |x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i x k / N} |k\rangle. \quad (\text{B.11})$$

Shor period finding: if $a^r \equiv 1 \pmod{N}$, factors from $\gcd(a^{r/2} \pm 1, N)$ when conditions hold.

B.6 CHSH and Protocols

CHSH parameter:

$$S = E(a, b) + E(a, b') + E(a', b) - E(a', b'), \quad |S| \leq 2 \text{ (LHV)}, \quad |S| \leq 2\sqrt{2} \text{ (QM)}. \quad (\text{B.12})$$

Teleportation cost: one ebit + two classical bits per unknown qubit.

Appendix C

Complexity Reference

This appendix records complexity classes and representative problem scalings. See Chapters 1 and 8 for context.

C.1 Growth Classes

Common asymptotic notations:

Notation	Example
$O(1)$	Array index lookup
$O(\log n)$	Binary search
$O(n)$	Linear scan
$O(n \log n)$	Merge sort
$O(n^2)$	Naive matrix multiply
$O(2^n)$	Exact TSP brute force

C.2 Complexity Classes

P: Decision problems solvable in polynomial time on a deterministic classical Turing machine.

NP: Decision problems whose yes-instances admit polynomial-size certificates verifiable in polynomial time.

BQP: Decision problems solvable in polynomial time on a quantum computer with error probability $< 1/3$, equivalently bounded-error quantum polynomial time.

Known containments: $P \subseteq BQP \subseteq PSPACE$. The relationship between **NP** and **BQP** is open; quantum computers are not known to solve NP-complete problems in polynomial time.

C.3 Algorithm Comparison Table

Table C.1: Representative query or time complexity (promise problems).

Problem	Classical	Quantum
Deutsch ($n = 1$)	2 queries (worst)	1 query
Deutsch–Jozsa	$2^{n-1} + 1$	1
Bernstein–Vazirani	n	1
Simon (period s)	$O(2^{n/2})$ queries	$O(n)$ queries
Grover unstructured search	$\Theta(N)$	$O(\sqrt{N})$
Quantum walk hits (typical)	varies	often quadratic speedups
Factoring (Shor, ideal)	no known poly-time	poly-time
Discrete log (Shor, ideal)	no known poly-time	poly-time

C.4 Simulation Complexity

- **General n -qubit state:** $O(2^n)$ amplitudes; exact simulation intractable for large n .
- **Clifford circuits:** Gottesman–Knill simulation in $O(n^2)$ per layer (polynomial).
- **Tensor networks:** structured states may simulate beyond 50 qubits with approximations.

C.5 Cautionary Notes

Different algorithms provide different advantages. Quantum computers are not universally exponentially faster. Always specify: problem family, promise conditions, metric (queries, time, samples), and classical baseline.

Appendix D

Glossary

Amplitude	Complex coefficient multiplying a basis ket in a superposition; probabilities derive from squared magnitudes.
Ancilla	Auxiliary qubit used temporarily during computation (e.g. oracle workspace, syndrome measurement).
Basis	Orthonormal set of vectors spanning a Hilbert space; measurement outcomes label basis elements.
Bit	Classical two-level information unit, 0 or 1.
Bloch sphere	Geometric representation of pure single-qubit states as points on the unit sphere, modulo global phase.
BQP	Bounded-error quantum polynomial time; decision problems efficiently solvable on a quantum computer.
Circuit depth	Number of sequential time layers in a quantum circuit, counting parallel gates on disjoint qubits as one layer.
Clifford gate	Gate generated by $\{H, S, \text{CNOT}\}$; simulable efficiently classically (Gottesman–Knill).
CNOT	Controlled-NOT gate; flips target when control is $ 1\rangle$.
Decoherence	Loss of quantum coherence through interaction with an uncontrolled environment.
Depolarizing channel	Noise model applying random I, X, Y, Z Pauli errors with fixed probability.
Ebit	One unit of shared entanglement, e.g. one Bell pair.
Entanglement	Non-separable quantum state of composite systems; cannot be written as a product of subsystem states.
Error syndrome	Classical bit pattern from stabilizer measurements indicating likely error location/type.

Fault tolerance

Computation scheme maintaining arbitrary low logical error rates given physical errors below a threshold.

Gate

Reversible operation on bits (classical) or unitary operation on qubits (quantum).

Global phase

Overall complex phase $e^{i\gamma}$ on a state vector; unobservable in standard measurements.

Hadamard gate (H)

Single-qubit gate exchanging Z and X bases: $H|0\rangle = |+\rangle$.

Hilbert space

Complex vector space with inner product where quantum states live.

Ket ($|\psi\rangle$)

Dirac notation for a column state vector.

Bra ($\langle\psi|$)

Conjugate transpose row vector dual to $|\psi\rangle$.

Logical qubit

Error-protected encoded qubit implemented across many physical qubits.

Magic state

Resource state enabling non-Clifford gates fault-tolerantly, especially T gates.

Measurement

Process yielding classical outcomes with Born-rule probabilities and state update.

Mitigation

Statistical techniques reducing noise bias without full quantum error correction.

NISQ

Noisy intermediate-scale quantum; devices too noisy for full fault tolerance yet useful for experiments.

No-cloning theorem

Unknown arbitrary quantum states cannot be copied by any unitary process.

No-signaling

Entanglement cannot transmit controllable information faster than light.

Oracle

Black-box unitary implementing a function in query complexity and algorithms.

Pauli operators

Single-qubit gates X, Y, Z ; building blocks for errors and stabilizers.

Phase

Argument of a complex amplitude; relative phase affects interference.

Physical qubit

Hardware-level two-level system subject to noise and calibration drift.

Post-quantum cryptography (PQC)

Classical cryptographic algorithms conjectured secure against quantum adversaries.

Projector

Operator $P = |m\rangle\langle m|$ associated with measurement outcome m .

Qubit

Two-level quantum information unit generalizing the classical bit.

Stabilizer

Pauli operator commuting with a code space; measured to extract syndromes.

Superposition

Linear combination of basis states with more than one nonzero amplitude.

Surface code Leading 2D topological quantum error-correcting code with local checks.

Tensor product

Composition of subsystem Hilbert spaces: $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$.

Threshold theorem

Result guaranteeing fault tolerance if physical gate error rates fall below a code threshold.

Unitary

Linear operator U with $U^\dagger U = I$; reversible quantum gate before measurement.

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